



KARAN RAJ T

AI/ML Engineer | LLM & RAG Systems

Sattur, Tamil Nadu | +91 9384102655 | karanraj2006rk@gmail.com

linkedin.com/in/karan-raj-t-835508351 | github.com/vip-sk07

AI and Data Science undergraduate specializing in Large Language Models, Retrieval-Augmented Generation, and applied machine learning. Builds local-first AI agents integrating multiple LLM providers, with hands-on experience in RAG pipelines, semantic search, agentic automation, and classical AI search algorithms.

TECHNICAL SKILLS

AI/ML: NumPy, Pandas, Scikit-learn, TensorFlow, PyTorch, Hugging Face, Ollama, RAG, Vector Databases, LLM Fine-Tuning

Languages: Python, C, C++, Core Java

Databases & Tools: MySQL, Oracle SQL, Git, GitHub, VS Code, Jupyter Notebook, Power BI

AREAS OF INTEREST

Large Language Models, Hugging Face, Llama.cpp, Retrieval-Augmented Generation, LLM Fine-Tuning, AI Model Optimization

EDUCATION

B.Tech, Artificial Intelligence and Data Science — Mepco Schlenk Engineering College, Sivakasi | Pre-Final Year (2024–2028) |

CGPA: 8.39/10

Higher Secondary Certificate (Class XII) — S.H.N. Edward Higher Secondary School, Sattur | 87.67% (2024)

Secondary School Leaving Certificate (Class X) — S.H.N. Edward Higher Secondary School, Sattur | 85.80% (2022)

PROJECTS

VIP Assistant — Local AI Agent with Advanced RAG

Jul 2026 – Present

Python, Node.js, React, Ollama, Hugging Face, RAG, Vector Database, WebSockets

- Built a local AI assistant integrating cloud and on-device LLMs with multi-provider support (Ollama, Gemini, Claude, NVIDIA NIM).
- Implemented Retrieval-Augmented Generation with semantic search and workspace indexing for intelligent code assistance.

Smart Beneficiary Mapping System

Feb 2026 – Present

Python, Flask, MySQL, AI Agents, Automation, Rule Engine

- Designed an AI-enhanced eligibility recommendation platform for government welfare schemes, evolving from a rule-based engine to an autonomous agent-driven system.
- Added autonomous agent workflows and intelligent recommendations to reduce manual effort and improve accessibility.

Stock Market Analysis & Prediction

Apr 2026 – May 2026

Python, Pandas, Scikit-learn, Streamlit, Machine Learning

- Built an interactive dashboard for stock market analysis and prediction using multiple ML algorithms, including regression, classification, and model comparison.

Maze Runner 3D using AI Search Techniques

Mar 2026 – May 2026

Python, Flask, JavaScript, Three.js, Artificial Intelligence

- Developed a 3D maze game demonstrating classical AI search algorithms with intelligent pathfinding and obstacle navigation.

CERTIFICATIONS

- Introduction to Internet of Things (Elite) — NPTEL, IIT Kharagpur | Jan–Apr 2026
- Design & Implementation of Human-Computer Interfaces — NPTEL, IIT Guwahati | Jul–Oct 2025
- IEEE English for Technical Professionals — IEEE | May 2025

LEADERSHIP & RESPONSIBILITIES

Google Student Club — Office Bearer

2025 – Present

The Institution of Engineers (India) — Executive Member

2025 – Present

AIMEE Association — Design and Publicity Member

2026 – Present

Scout — Scout Leader

2020 – 2022

MERIT & RECOGNITION

- Certificate of Merit — Executive Member, The Institution of Engineers (India) – AI & DS Student Chapter (2025–2026).
- Certificate of Merit — Office Bearer, Google Student Club (2025–2026).
- Presented Smart Beneficiary Mapping Platform at TECHNOINNOVATE 2025–2026, IIC, Mepco Schlenk Engineering College.

ADDITIONAL INFORMATION

Coding Profiles: LeetCode, HackerRank, CodeChef | Languages: English, Tamil